Report on Proposed Minimum Order Quantity (MOQ) for Finalized Bacterial Biomarkers

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Date: December 1, 2025
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Project: Nano-technology enhanced biosensor for early bacterial sepsis detection

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Executive Summary

This report presents the proposed minimum order quantities (MOQ) for three prioritized bacterial biomarkers: N-acetylmuramic acid (NAM), Lipopolysaccharides (LPS), and Lipoteichoic acid (LTA). These markers enable universal bacterial detection with simultaneous Gram classification using Surface-Enhanced Raman Spectroscopy (SERS) methodology. The procurement plan adheres to IIT Jodhpur's Stores and Purchase Section (SPS) guidelines and General Financial Rules (GFR) 2017[1][2].

Biomarker Panel Rationale:

- **NAM (Universal Bacterial Detection):** Present in 100% of bacteria, absent in viruses, fungi, and parasites[3][4]
- **LTA (Gram-Positive Classification):** Unique to Gram-positive bacterial cell walls; 10-30% of cell wall mass[5]
- **LPS (Gram-Negative Classification):** Exclusive to Gram-negative bacteria; primary endotoxin mediator[6][7]

Proposed Minimum Order Quantities (MOQ)

3.2 Proposed MOQ Table

Biomarker	**Supplier Form**	**MOQ**	**Unit**	**Est. Cost (INR)**
N-Acetylmuramic Acid (NAM)	Analytical standard, lyophilized	100	mg	₹12,000-20,000
Lipopolysaccharide (LPS, E. coli O111:B4)	Standard endotoxin, lyophilized	10	mg	₹12,000-18,000
Lipoteichoic Acid (LTA, Gram-positive extract)	Purified antigen, solution/lyophilized	10	mg	₹30,000-40,000
Total MOQ Procurement	**₹54,000-78,000**			
Contingency (15%)	**₹12,000-20,000**			
Total Recommended Budget	**₹66,000-98,000**			

3.3 Justification for MOQ Quantities

- NAM: 0.5-1 mg per experiment (100 mg = 100-200 experiments)
- LPS: 0.1 mg per experiment (10 mg = 100 experiments)
- LTA: 0.05 mg per experiment (5 mg = 100 experiments)

Vendors sell complete vials, so buying packages in mg. The cost efficiency is maintained

5. Vendor Requirements and Technical Specifications

5.1 Vendor Selection Criteria

1. **Manufacturing and Quality Standards:**
    - ISO 9001:2015 certification (quality management)
    - ISO/IEC 17025:2017 (analytical testing, if applicable)
    - GMP certification for biochemical reagent production
    - Experience supplying research-grade biomarkers to academic/biotech institutions
2. **Technical Capabilities:**
    - Proven supply of NAM, LPS, or LTA to peer research groups
    - Ability to supply materials with specifiable purity and endotoxin levels
3. **Quality Assurance Documentation:**
    - Certificate of Analysis (CoA) for EACH batch (mandatory)
    - Purity: ≥95% for NAM; ≥90% for LPS/LTA solutions
    - Endotoxin levels: <0.25 EU/µg (for pyrogenicity/SERS compatibility)
    - Identity confirmation via HPLC, MS, or NMR
    - Stability data: Minimum 24-month shelf-life at specified storage (2-8°C or -20°C)
4. **Delivery and Support:**
    - Delivery within 2-4 weeks from PO date
    - Cold-chain packaging (4°C or -20°C as required)
    - Technical documentation: MSDS, storage protocols, usage guidance
    - Post-delivery technical support (troubleshooting, handling advice)

5. **Regulatory Compliance:**
    - GST registration (Indian suppliers) / FEMA compliance (imports)
    - If imported: Valid import license, Customs clearance commitment
    - Compliance with Indian Standards (BIS) or equivalent

5.2 RFQ Technical Specifications Checklist

For NAM:

- Molecular formula: $C_8H_{13}NO_5$ (MW 219.2 g/mol); verify via mass spectrometry
- Purity: ≥98% by HPLC
- Physical form: Lyophilized powder (white to off-white)
- Storage: -20°C, protected from light
- Shelf life: ≥24 months from manufacture

For LPS (E. coli O111:B4):

- Molecular identity: Confirmed endotoxin from *E. coli* O111:B4 strain
- Endotoxin activity: 2,000-4,000 EU/µg (USP reference grade preferred)
- Pyrogenicity: Passes rabbit pyrogenicity test or LAL (Limulus Amebocyte Lysate) test
- Purity: ≥90% by HPLC
- Physical form: Lyophilized powder or 1 mg/mL solution (specify)
- Storage: 2-8°C or -20°C (per supplier recommendation)

For LTA (Gram-positive source—recommend *S. aureus or B. subtilis):*

- Biological source: Specify bacterial strain and extraction method
- Purity: ≥90% by HPLC or electrophoresis
- Molecular mass: Consistent with polyglycerol-phosphate polymer (typically 5-50 kDa)
- Identity: Western blot or antibody confirmation (provide protocol)
- Endotoxin levels: <0.5 EU/µg (confirm absence of LPS contamination for Gram-positive discrimination)
- Physical form: Purified antigen, lyophilized powder or solution
- Storage: 2-8°C or -20°C

All Biomarkers—Common Requirements:

- Complete CoA with batch number, manufacture date, expiration date
- MSDS (Material Safety Data Sheet) in English
- Lot traceability documentation
- Warranty: 100% refund if CoA specifications not met upon receipt
- Partial shipment policy: Allow phased delivery if necessary

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